

Expt 3 Performance Analysis of IIR Filters

3.1 Design and Comparative Analysis of Butterworth and Chebyshev Type-I IIR Band-Pass Filters Using MATLAB

```
1  clc; clear; close all;
2  % Specifications
3  Fs=2000; Rp=1; Rs=40;
4  Wp=[300 600]/(Fs/2);
5  Ws=[200 700]/(Fs/2);
6
7  % Butterworth Filter
8  [n1,M1]=buttord(Wp,Ws,Rp,Rs);
9  [b1,a1]=butter(n1,M1,'bandpass');
10
11 % Chebyshev Type-I Filter
12 [n2,M2]=cheb1ord(Wp,Ws,Rp,Rs);
13 [b2,a2]=cheby1(n2,M2,'bandpass');
14
15 % Magnitude Response
16 figure;
17 freqz(b1,a1); title('Butterworth Filter');
18
19 figure;
20 freqz(b2,a2); title('Chebyshev Type-I Filter');
21
22 % Group Delay
23 figure;
24 grpdelay(b1,a1); hold on;
25 grpdelay(b2,a2);
26 legend('Butterworth','Chebyshev-I');
27
28 % Pole-Zero Plot
29 figure;
30 subplot(121),zplane(b1,a1),title('Butterworth')
31 subplot(122),zplane(b2,a2),title('Chebyshev-I')
32
33 % Impulse Response
34 figure;
35 subplot(211),impz(b1,a1),title('Butterworth Impulse')
36 subplot(212),impz(b2,a2),title('Chebyshev Impulse')
37
38 % Pole Magnitude
39 figure;
40 subplot(121),stem(abs(roots(a1)),'filled')
41 title('Butterworth Pole Magnitude')
42
43 subplot(122),stem(abs(roots(a2)),'filled')
44 title('Chebyshev Pole Magnitude')
45
```

Command Window

New to MATLAB? See resources for [Getting Started](#).

Butterworth Order = 9
Chebyshev Order = 5
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